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Advanced Apex Project II

Week 2 — Project Outline Report

Case Study: Employee Attrition in Retail

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Programme	M.Sc. Data Science & Artificial Intelligence
Submission	Week 2 — Project Outline & Initial Setup

1. Problem Statement & Business Goal

Employee turnover is a critical operational and financial challenge in the retail industry. High attrition rates — particularly among front-line roles such as Cashiers and Sales Associates — lead to significant recruitment and training costs, loss of operational consistency, and declining customer satisfaction.

Business Goals

- Identify the key drivers of voluntary employee turnover in a retail context.
- Build and compare multiple classification models to predict employee attrition risk.
- Provide actionable recommendations to store management to improve staff retention.

2. Dataset Details

Dataset: Employee_Dataset_Retail.csv (Authored by Manu Chopra, available on Kaggle).

Property	Value
Total Records	9,000 employees
Total Features	31 columns
Target Variable	Attrition_Status (Yes / No)
Attrition = Yes	655 employees (7.28 %)
Attrition = No	8,345 employees (92.72 %)
Class Imbalance	≈ 12.7 : 1 (No : Yes)

Feature Dictionary (Key Features)

Feature	Description	Data Type
Employee_ID	Unique employee identifier — dropped before modelling	Numeric (int)
Employee_Name	Employee name — dropped before modelling	Categorical (ID)
Gender	Male / Female	Categorical
Age	Employee age (22–59 years)	Numeric (int)
Date_of_Joining	Date employee joined the company	Date (string)
Years_at_Company	Tenure at company in years (1–9)	Numeric (int)
Monthly_Income	Monthly salary in INR (₹20,010–₹1,49,985) — no missing values	Numeric (int)
Annual_Income	Yearly salary in INR	Numeric (int)
Department	Finance, HR, Sales, Marketing, IT, Operations, Supply Chain, Customer Support (8 depts)	Categorical
Job_Role	Assistant, Executive, Specialist, Supervisor, Analyst, Technician, Manager, Coordinator	Categorical
Position	Seniority level — Junior, Mid-Level, Senior, Lead	Ordinal
Employment_Type	Full-time, Contract, Part-time	Categorical
Work_Location	HQ or Remote	Binary
State	10 Indian states (Maharashtra, Gujarat, Tamil Nadu, etc.)	Categorical
City	50 anonymised cities (City_1...City_50)	Categorical
Distance_from_Home	Commute distance in km (1–49)	Numeric (int)
Education_Level	High School, Diploma, Graduate, Postgraduate, PhD	Ordinal
Education_Field	Engineering, Commerce, Arts, Management, Science, Others	Categorical
Marital_Status	Single, Married, Divorced	Categorical
Performance_Score	Manager-rated grade — A*, A+, A, B	Ordinal

Feature	Description	Data Type
Last_3_Year_Rating	Multi-year performance label (9 categories: Future Star, Contributor, etc.)	Categorical
Attrition_Status	TARGET — Employee left (Yes / No)	Binary
Attrition_Reason	Specific reason for leaving — NaN for retained employees (exclude before model training)	Categorical
Attrition_Date	Date of departure — NaN for retained employees (exclude before model training)	Date (string)
Onboarding_Status	Accepted, Rejected, Pending, Rolled out	Categorical
New_Hires_Joining_Month	Month number of joining (1–12)	Numeric (int)
Environment_Satisfaction	Workplace satisfaction — Very High, High, Medium, Low	Ordinal
Employee_Cost_to_Company	Total annual CTC in INR (₹5L–₹25L)	Numeric (int)
Hourly_Rate	Hourly pay rate (₹200–₹999)	Numeric (int)
Annual_HR_Feedback_Satisfaction_Score	Annual HR feedback satisfaction score (1–4)	Ordinal
Yearly_Total_Training_Hours	Total training hours per year (10–199)	Numeric (int)

3. Initial Dataset Understanding

Observations

- **Data Volume:** 9,000 records — sufficient for reliable model training and validation.
- **Data Types:** Features are a mix of numerical, categorical, and ordinal types. Date_of_Joining is object type and needs datetime conversion.
- **Missing Values: Only two columns have missing values — Attrition_Reason and Attrition_Date — both intentionally NaN for retained employees (not a data quality issue).**
 - No genuine missing values in predictor columns. Attrition_Reason and Attrition_Date are intentionally NaN for non-attrited employees and will be excluded from model training.
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 - Annual_HR_Feedback_Satisfaction_Score: no missing values — column is complete in the new dataset.
 - Environment_Satisfaction: no missing values — column is complete in the new dataset.
 - Attrition_Reason: 8,345 missing (92.7%) and Attrition_Date: 8,345 missing (92.7%) — intentionally NaN for retained employees only; excluded from model training to prevent target leakage.
- **Class Imbalance:** 92.72 % No vs 7.28 % Yes (12.7 : 1 ratio). F1-Score and ROC-AUC are preferred over plain accuracy.

Expected Challenges

- Class imbalance must be handled via SMOTE or class_weight='balanced'.
- Severe class imbalance (12.7:1) requires SMOTE or class_weight='balanced' carefully tuned to avoid over-sampling noise.
- Date_of_Joining requires datetime conversion and feature extraction (Joining_Year, Joining_Month).
- Ordinal encoding must preserve natural ordering for Position, Education_Level, Performance_Score, and Environment_Satisfaction.

4. Proposed Workflow

Step 1 — Data Preprocessing (DPP)

- Drop Employee_ID (non-predictive identifier).
- Parse Date_of_Joining to datetime; extract Joining_Year and Joining_Month.
- Verify data integrity: confirm no missing values in numeric columns (new dataset has none in predictors).
- Drop Employee_Name (non-predictive; all values are anonymised as 'Employee_N').
- Encode Onboarding_Status (4 categories: Accepted, Rejected, Pending, Rolled out) using One-Hot Encoding.

Step 2 — Exploratory Data Analysis (EDA)

- Univariate analysis: distributions of Age, Monthly_Income, Years_at_Company, Distance_from_Home, Yearly_Total_Training_Hours.
- Bivariate analysis: Attrition_Status vs Employment_Type, Department, Job_Role, Work_Location, Position, Distance_from_Home.
- Correlation heatmap of all numeric features.
- Attrition_Status rate by Department, State, and Work_Location (grouped bar charts and heatmaps).

Step 3 — Feature Engineering

- IncomeToRoleRatio: $\text{Monthly_Income} / \text{median income of the same Job_Role}$.
- SalaryPerYear: $\text{Monthly_Income} \times 12 / \text{Employee_Cost_to_Company}$ — captures salary-to-CTC efficiency ratio as a retention risk signal.
- TenureBand: bin Years_at_Company into groups (1–3, 4–6, 7–9) aligned with the dataset range of 1–9 years.

Step 4 — Encoding & Scaling

- One-Hot Encode nominals: Gender, Department, Job_Role, Employment_Type, Work_Location, Education_Field, Marital_Status, State, Onboarding_Status, Last_3_Year_Rating.
- Ordinal Encode: Education_Level (High School < Diploma < Graduate < Postgraduate < PhD), Position (Junior < Mid-Level < Senior < Lead), Performance_Score (B < A < A+ < A*), Environment_Satisfaction (Low < Medium < High < Very High), Annual_HR_Feedback_Satisfaction_Score (1–4 already numeric).
- StandardScaler on all numeric features.

Step 5 — Class Imbalance Handling

- Apply SMOTE only on training data using imblearn.pipeline (prevents data leakage).
- Alternative: class_weight='balanced' for tree-based models.

Step 6 — Model Building & Evaluation

- Train/test split: 80 % train, 20 % test (stratified by target).
- Train all 5 proposed models and record metrics.
- Evaluate using F1-Score (Yes class), ROC-AUC, Precision, Recall, Confusion Matrix.

Step 7 — Hyperparameter Tuning

- GridSearchCV for Logistic Regression and Decision Tree.

- RandomizedSearchCV for Random Forest and Gradient Boosting.
- 5-fold Stratified K-Fold cross-validation throughout.

5. Proposed Baseline & Advanced Models

Model	Type	Rationale
Logistic Regression	Baseline	High interpretability; good linear decision boundary
Decision Tree	Advanced	Captures non-linear relationships; feature importance
Random Forest	Advanced	Ensemble of trees; robust to overfitting
Gradient Boosting	Advanced	Typically best performer on tabular classification
K-Nearest Neighbors	Advanced	Distance-based comparison model

6. Evaluation Metrics & Tuning Plan

Evaluation Metrics

Metric	Why Chosen
F1-Score (Yes class)	Balances Precision & Recall under class imbalance
ROC-AUC	Discriminative ability across all decision thresholds
Precision	Avoids unnecessary HR interventions for stable employees
Recall	Ensures high-risk employees are not missed
Confusion Matrix	Visual TP / FP / TN / FN breakdown

Tuning Strategy

- Logistic Regression: C, solver, max_iter — GridSearchCV.
- Decision Tree: max_depth, min_samples_split, criterion — GridSearchCV.
- Random Forest: n_estimators, max_depth, min_samples_leaf — RandomizedSearchCV.
- Gradient Boosting: n_estimators, learning_rate, max_depth — RandomizedSearchCV.
- All tuning uses 5-fold Stratified K-Fold cross-validation.

7. Tools & Technologies

Tool / Library	Version	Purpose
Python	3.12.12	Core programming language
pandas	3.0.3	Data loading, manipulation, and inspection
numpy	2.4.6	Numerical operations and array processing
matplotlib	3.10.9	Data visualization and plotting
seaborn	0.13.2	Statistical visualization
scikit-learn	1.8.0	Preprocessing, models, evaluation, pipelines
imbalanced-learn	0.14.1	SMOTE for handling class imbalance
Jupyter Notebook	7.5.6	Interactive development and documentation

8. References

- Employee_Dataset_Retail by Manu Chopra (Kaggle). URL: <https://www.kaggle.com/datasets/carebymanu/employee-dataset-retail> (Note: Dataset link may have been updated, but authored by Manu Chopra).
- BITS Pilani M.Sc. DS & AI Course Materials — Advanced Apex Project II.
- Scikit-learn Documentation: <https://scikit-learn.org/stable/>
- imbalanced-learn (SMOTE): <https://imbalanced-learn.org/stable/>
- Kuhn, M., & Johnson, K. (2013). Applied Predictive Modeling. Springer.